

Curriculum vitae

Applied mathematics student in Norway, working on machine learning, computational geometry and systems programming. Looking for a junior developer or data role where the work is technical.

EMAIL	PHONE	LOCATION	GITHUB
ulyanagrab333@gmail.com	+47 46 55 57 59	Norway	github.com/draurug

SELECTED WORK

Longer write-ups of each project are on the work page.

Ethereum Fraud Detection 2026

Classified 9,841 Ethereum addresses as fraudulent or legitimate from 50 transaction-derived features. LightGBM reached F1 0.987 on held-out data. Found and fixed a leak first: a naive random split put 5.7% of test rows in the training set through duplicate feature profiles, so the split was made group-aware and models were selected by cross-validated PR-AUC rather than by test score.
Python · scikit-learn · LightGBM · XGBoost · pytest

CrossW 2026

Designed and shipped a themed crossword site, from the product idea through to a live deployment. Four content packs, 75 unit tests, roughly 5,400 lines of TypeScript. Answers are compiled to opaque indices at build time so they never reach the browser in readable form.
Next.js 15 · React 19 · TypeScript · Vitest · Cloudflare Workers

Voronoi Path Planner 2026

Interactive path planner that optimises for clearance rather than distance: it builds a clearance field against obstacles, binary-searches the largest threshold that keeps start and goal connected, then runs Dijkstra inside that corridor. Accompanied by a generated figure set and a written report.
C++ · OpenGL · FreeGLUT · NumPy · SciPy

Networked Tic-Tac-Toe and Battleship 2026

Two Qt desktop games in C++. The first plays over TCP with a packet protocol on Boost.Asio, an invitation handshake and turn synchronisation between clients; the second keeps the game rules entirely free of Qt so the widget only paints.
C++17/20 · Qt · Boost.Asio · CMake

TECHNICAL SKILLS

LANGUAGES	DATA AND MACHINE LEARNING
Python, C++ (17/20), TypeScript, JavaScript, SQL, C#, Octave/MATLAB, LaTeX	scikit-learn, LightGBM, XGBoost, pandas, NumPy, SciPy, matplotlib, seaborn, Jupyter, Power BI
WEB AND TOOLING	OTHER
Next.js, React, Vue, Node, PostgreSQL, Git, CMake, pytest, Vitest, Docker, Cloudflare Workers	OpenGL, Qt, Boost.Asio, Unity and C#, Blender, Arduino

EDUCATION

BSc Applied Mathematics — Odesa I. I. Mechnikov National University 2023 — present

Institute for Optimal Control and Economic Cybernetics. Coursework in numerical methods, computational geometry and computer graphics, decision theory, blockchain technology and machine learning. Course paper on fraud detection, plus a bonus project on Voronoi-based path planning.

IT Step Academy — IT-focused secondary programme 2019 — 2022

Programming fundamentals, game development in Unity and C#, 3D modelling in Blender, and Arduino.

Odesa Lyceum no. 1 (A. P. Bystrina) — secondary education 2016 — 2023

English specialisation.

EXPERIENCE

Logistics manager — SmartAuto LLC May — Aug 2024

Coordinated vehicle logistics and kept the operational records that the scheduling depended on.

Mentor — IT2School

Dec 2023 — Jan 2025

Taught programming to school students and supported them through their assignments.

Project lead — Mechnykov student card

2023 — present

Led the student card project at the university. Member of the Student Union and of AtomSpace.

LANGUAGES

- **Ukrainian** and **Russian** — native
 - **English** — C1, certified by the London School of English (2025)
 - **Norwegian** — B2 course completed, Læringssenteret for voksne i Ringerike kommune; Norskprøven passed
 - **German** — Goethe-Institut certificate
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CERTIFICATES

- English C1 — London School of English, 2025
 - Working with Data in Python — DataCamp, 2024
 - Data Visualisation in Power BI — DataCamp, 2024
 - Data Analysis in Google Sheets — DataCamp, 2024
 - C++ Intermediate — Sololearn, 2023
 - C++ Basics — Coursera, 2023
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REFERENCES

Available on request.